

Regs Checker — Data & Taxonomy Reference

Audience: Business, product, and review teams (no engineering background assumed)

Purpose: Explains, in plain English, every field the pipeline extracts from each law and how the current taxonomy is organized.

Authoritative law count: 232 (`data/fact_laws.csv`)

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1. The big picture

Regs Checker reads the **full text of ~232 US state and federal AI laws** and turns each one into **structured, queryable data** — so that instead of a lawyer reading a 40-page bill, a business can ask *"Does this law apply to me? What do I have to do? By when? What happens if I don't?"* and get a precise answer.

It captures data at **two levels**:

Level	Question it answers	How it's produced
Law-level (one record per law)	"What is this law, who does it cover, what's the penalty, when's the deadline?"	3 AI agents that read the <i>whole bill</i> at once
Clause-level (many records per law)	"What exactly does <i>this sentence</i> require, and of whom?"	6 AI agents that read the bill <i>passage by passage</i>

Every extracted fact is tied back to a **verbatim quote** from the law (the "evidence span") and graded for **trustworthiness** (a confidence tier from A to D). Nothing is taken on faith.

2. Law-level fields

These are the headline facts held for each law — exported per-law to `output/laws.json` . They come partly from source metadata (Orrick law-firm tracker + IAPP) and partly from the three whole-bill AI agents. (*A consumer-facing "Law Card" UI built on this data is planned, not yet built* — see Section 6.)

2a. Basic identity & status

Field	Plain-English meaning	Example
Title	The law's name	"Colorado AI Act (SB 205)"
Bill number	Official legislative number	"SB 205"
Jurisdiction	Which state / body	"Colorado", "California", "EU"
Status	Where it is in its lifecycle	<i>Proposed -> Passed, not yet in effect -> In effect</i>
Effective date	When it becomes (or became) law	2026-02-01
AI scope summary	One-line topic	"Automated decision-making in employment"
Source	Where we sourced the law	Orrick, IAPP

2b. "Who does this law apply to?" — *Applicability Agent*

The engine behind the "does this apply to me?" matching. Reads the whole bill and answers:

Field	Plain-English meaning	Example values
Covered entity types	The kinds of businesses regulated	developer, deployer, provider, employer, state agency
Covered sectors	Industries in scope	employment, healthcare, credit, housing, insurance, criminal justice
AI system types in scope	What kinds of AI are regulated	high-risk AI, automated decision systems, generative AI, facial recognition
Size thresholds	The "you're big enough to be covered" triggers	revenue >= \$25M; >= 50 employees; data on >= 100k consumers; compute >= 10 ²⁶ FLOPS
Geographic scope	Where it reaches	"entities doing business in Colorado"
Key exemptions	Who's carved out	"small businesses under 50 employees", "HIPAA-covered entities"
Government only?	Applies only to the public sector?	true / false
Applicability summary	Plain-language "who's covered"	"Applies to businesses deploying AI for hiring decisions in CA."

2c. "What happens if I don't comply?" — *Enforcement Agent*

Field	Plain-English meaning	Example
Enforcing body	Who comes after you	"Attorney General"
Max civil penalty (USD)	Biggest fine per violation	\$20,000
Penalty per	The unit fines are counted in	per violation / per day / per occurrence
Cure period (days)	Grace period to fix it before penalties	60 days
Private right of action	Can individuals sue you directly?	true / false
Criminal penalties	Possible jail time / criminal fines?	true / false
Criminal penalty description	Detail of any criminal penalties	"Class 1 misdemeanor; up to 1 year"
Enforcement text	Verbatim quote describing the enforcement mechanism	"...shall be enforced by the attorney general..."

2d. "When are the deadlines?" — *Compliance Timeline Agent*

Field	Plain-English meaning	Example
Law effective date	When the law switches on	2026-02-01
Enforcement start date	When they actually start penalizing	2026-08-01
Sunset date	When the law expires (if ever)	—
Key deadlines	List of dated actions, each with: action, deadline type (before-deployment / after-enactment / recurring / event-triggered / one-time), relative days, recurrence months, trigger event	"Complete impact assessment 90 days before deployment"
Assessment frequency	How often you must re-do assessments	every 12 months
Consumer response window	Days to respond to a consumer request	45 days
First compliance action	The first thing you must do, and when	"Register the system with the AG before launch."

3. Clause-level fields (the detailed extractions)

These six agents read the law **passage by passage**, producing many records per law. This is the granular layer that powers detailed obligation tracking and cross-state comparison.

Stored with every extraction record (all 9 agents, bill- and clause-level). Each record also carries: **evidence spans** (verbatim backing quotes, each with a *verified* flag + character offsets), a **confidence score + tier** (A–D, Section 5), a **review status** (pending / approved / rejected / needs-revision), the **model + prompt version** used, and a deterministic **plain-English summary** for reviewers.

3a. "Who must do what?" — *Obligation Agent* (the richest extractor)

Field	Plain-English meaning	Example
Subject / Subject (normalized)	Who must comply (raw + standardized)	"a deployer of a high-risk system" -> deployer
Modality	The strength of the duty	must / shall / may / prohibited
Action	What they must do (or not do)	"conduct an impact assessment"
Object	What the action applies to	"the automated decision system"
Condition	The trigger	"before deploying the system"
Timeline	Dates attached to this duty	deadline, phase-in period, sunset
Enforcement	Penalty attached to this duty	body, penalty type, max fine, cure period, can-be-sued flag
Preemption signals	Verbatim federal/state preemption language found in the clause	"this section does not preempt federal law"
Section reference	Where in the law this clause sits	"§ 6-1-1702(3)(a)"
Jurisdiction	Jurisdiction code if stated in the clause	"CO"
Safe harbor	A "get-out-of-jail" path	"Following NIST AI RMF = affirmative defense"
Consent requirements	Required notice/consent	type (opt-in/opt-out), timing, method
Interpretation risks	Vague/ambiguous language flagged for review	"'reasonable care' is undefined" (severity: high)

3b. "Who is protected, and what can they demand?" — *Rights & Protections Agent*

The flip side of obligations — what *individuals* are entitled to.

Field	Plain-English meaning	Example
Right holder	Who holds the right	consumer, employee, job applicant, patient
Right holder (normalized)	Standardized holder category	consumer / employee / public
Protected categories	Explicitly protected groups	minor, tenant, borrower, student
Right type	The kind of right	notice, explanation, opt-out, appeal, deletion, human review
Right description	What they're entitled to	"right to a human review of an adverse AI decision"
Trigger condition	When the right kicks in	"upon an adverse hiring decision"
Duty bearer	Who must honor it	the employer / deployer
Remedies	Recourse if violated	complaint, damages, deletion — plus time limits
Section reference / Jurisdiction	Where in the law, and jurisdiction code	"§ 22-...", "CA"
Interpretation risks	Same ambiguity flags as obligations (vague terms, undefined refs)	"'meaningful human review' is undefined"

3c. "How do I prove compliance?" — *Compliance Mechanism Agent*

The procedural machinery: audits, assessments, reporting, recordkeeping.

Field	Plain-English meaning	Example
Mechanism type	The procedural requirement	impact assessment, bias audit, registration, certification, reporting
Responsible party	Who performs it	developer / deployer

Field	Plain-English meaning	Example
Responsible party (normalized)	Standardized category	developer / deployer / operator / vendor
Audits	Audit details	type, frequency, who audits, who sees results, public?
Record retention	How long to keep records	36 months — and <i>what</i> to keep
Assessment frequency	How often impact assessments must be renewed (months)	12 (annual)
Reporting	Filing cadence + recipient	annual report to the AG
Classification flags	Quick yes/no tags	is bias testing? / is red-teaming? / is third-party audit?
Incident reporting window	Hours to report an incident	72 hours
NIST references	Links to NIST AI framework controls	"MEASURE-2.1"
Section reference / Jurisdiction	Where in the law, and jurisdiction code	"§ 1798.185", "CA"

3d. "When does it apply / when am I exempt?" — *Threshold & Exception Agent*

Field	Plain-English meaning	Example
Threshold sub-type	The kind of boundary	scope (who/what), temporal (deadlines), exemption (carve-outs)
Threshold type	The specific type within the sub-type	numeric / monetary / date / compute / entity_type / sector / safe_harbor
Threshold value / unit / condition	The actual trigger	"\$25,000,000 annual revenue"
Applies to obligation	Which obligation this boundary modifies	"the impact-assessment duty"
Revenue / employee / consumer-data thresholds	Typed numeric triggers	>= \$25M / >= 50 / >= 100k
Compute threshold (FLOPS) + description	Frontier-model size trigger, numeric + readable	10 ²⁶ — "models trained above 10 ²⁶ FLOPS"
Sector applicability	Which consequential-decision sectors	healthcare, employment, credit
Exceptions	Carve-outs and safe harbors	"research use is exempt"

3e. "What do the words mean?" — *Definition & Actor Agent*

Field	Plain-English meaning	Example
Term + Definition text	A defined term and its full legal definition	"'Algorithmic discrimination' means..."
Scope	Where the definition applies	"for purposes of this article"
Cross-references	Other sections that reference this term	"§ 2, § 7(b)"
Actors	Roles named, and their responsibilities	developer, deployer, regulator
Framework references	External standards the law leans on	"incorporates NIST AI RMF"

3f. "Does this conflict with other laws?" — *Preemption Agent*

Field	Plain-English meaning	Example
Conflict type	The kind of legal tension	federal preemption, cross-state conflict, First Amendment
Description	Plain-language conflict summary	"May be preempted by federal AI executive order"
Related authority	The competing law/authority	"Dec 2025 Federal EO on AI"
Severity	Risk level	high / medium / low
Preemption language	Verbatim preemption clause from the passage	"nothing in this section shall preempt federal law"
Cross-law references	Other laws this one points to	supersedes / incorporates / conflicts-with
Section reference / Jurisdiction	Where in the law, and jurisdiction code	"§ 4", "TX"

4. The current taxonomy (controlled vocabularies)

The taxonomy is the set of **standardized labels** that make the data filterable and comparable across states. Without it, "deployer," "Deployer," and "entity that deploys" would all be different — and filtering would break. Today the taxonomy lives in a handful of **dimension tables** (the "approved value lists") plus **mapping tables** (which law has which labels).

Dimension tables (the approved value lists)

Taxonomy	Approved values	Purpose
Jurisdictions	49 entries — all US states + EU	Filter by location
Actor types	Deployer, Developer, Provider, Distributor	Who the law regulates
Requirement types	12 values: Governance Program, Assessments, Training, Responsible Individual, General Notice, Labeling/Notification, Explanation/Incident Reporting, Provider Documentation, Registration, Third-party Review, Opt-out/Appeal, Nondiscrimination	The category of obligation
AI scopes	A = all AI systems · F = foundation/frontier models · D = automated decision-making · G = generative AI · * = AI trained on personal data	What kind of AI is covered
Legislative statuses	Active, Enacted, Failed/Dead, Signed	Lifecycle stage
Sources	Orrick, IAPP	Where the law data came from

Mapping tables (which law has which labels)

- **Law -> Requirements** — links each law to its requirement types *and* the responsible actor, with a mandatory-or-not flag.
- **Law -> AI scopes** — links each law to its scope code(s).

Tag categories (the search/filter facets in the `laws.json` export)

The user-facing export also groups tags into facets: **jurisdiction**, **source**, **lifecycle status**, **AI scope**, **AI topic**, **concept**, **compliance requirement**, **regulated actor** — e.g. concepts like *children*, *employment*, *intimate images*, *political advertising*, *transparency*.

5. How data quality is graded (the trust layer)

Because the data is AI-extracted, every record carries a **confidence tier** so reviewers and downstream products know how much to trust it. The score blends six signals — most heavily, **agreement with Orrick law-firm reference data (30%)** and **independent re-checking (25%)**.

Tier	Score	What it means for the business
A	≥ 0.85	High confidence — candidate for auto-approval
B	≥ 0.70	Solid — standard review
C	≥ 0.50	Needs a careful human look
D	< 0.50	Requires human review

The key rule (the "Orrick gate"): any extraction *without* law-firm reference data to validate against is automatically **Tier D**, no matter how good it looks. This is the guardrail that keeps unvalidated AI output from reaching production.

6. Where the taxonomy is heading

The taxonomy is mid-redesign. The strategy work underway expands it toward richer, profile-matched dimensions — `covered_sectors`, a two-level `obligation_type`, `harm_categories` (discrimination, privacy, deception, safety, child, election...), and split status fields (signed vs. actually-in-effect). The redesign principle is *additive* — old labels stay until new ones are validated.

If the business team will rely on these filters, it's worth knowing which dimensions are **live today** (Section 4) vs. **planned**. See `docs/taxonomy_strategy_summary.md` and `docs/taxonomy_dev_plan.md` for the full redesign plan, and `docs/pipeline_rebuild_plan.md` for the alternative ground-up approach.

7. Appendix — Taxonomy chart (for a legal data scientist)

This section is the schema-accurate companion to Section 4. It enumerates every controlled vocabulary as *it actually exists in the code today*, with its **enforcement level** and **join keys**, so the data can be queried and trusted appropriately.

The "taxonomy" is not one thing — it lives in **three coexisting layers** with very different reliability guarantees:

Layer	What it classifies	Where it lives	Enforcement
L1 — Curated law-metadata star schema	Whole laws	<code>data/*.csv</code> -> seeded to Postgres/Supabase	Hard — PK/FK, hand-curated
L2 — Operational Postgres enums	Pipeline state + extraction record type	DB <code>CREATE TYPE</code> enums	Hard — DB-rejected if off-list
L3 — Extraction vocabularies	The obligations/rights/etc. inside laws	Agent prompts + Pydantic schemas	Advisory — LLM-emitted, mostly unvalidated

The single most important fact for analysis: **L3 is almost entirely advisory**. Agents are *instructed* to emit values from a list, but the Pydantic schemas type these as plain `str` — so off-vocab values pass validation and land in `payload JSONB` unchanged. Only two L3 fields are schema-enforced (`Literal` -typed). This gap is the reason the normalization stage in the taxonomy redesign exists.

7.1 L1 — Curated law-metadata star schema

Entity-relationship diagram (Mermaid source — renders as a diagram in Markdown viewers):

```
erDiagram
    fact_laws ||--o{ dim_jurisdictions : jurisdiction_id
    fact_laws ||--o{ dim_legislative_statuses : status_id
    fact_laws ||--o{ dim_sources : source_id
    fact_laws ||--o{ map_law_requirements : law_id
    map_law_requirements ||--o{ dim_requirement_types : req_type_id
    map_law_requirements ||--o{ dim_actor_types : actor_id
    fact_laws ||--o{ map_law_scopes : law_id
    map_law_scopes ||--o{ dim_ai_scopes : scope_code
    fact_laws {
        int law_id PK
        string canonical_law_id
        string bill_number
        int jurisdiction_id FK
        int status_id FK
        date effective_date
        string title
        string ai_scope_summary
        text key_requirements_raw
        text enforcement_penalties
        int source_id FK
        string source_url
        string iapp_scope
        string iapp_section
    }
    map_law_requirements {
        int map_id PK
        int law_id FK
        int req_type_id FK
        int actor_id FK
        bool is_mandatory
        text compliance_notes
    }
    map_law_scopes {
```

```

    int map_id PK
    int law_id FK
    string scope_code FK
}
dim_jurisdictions {
    int jurisdiction_id PK
    string name
    string state_abbrev
}
dim_legislative_statuses {
    int status_id PK
    string status_name
}
dim_sources {
    int source_id PK
    string source_name
}
dim_requirement_types {
    int req_type_id PK
    string requirement_name
}
dim_actor_types {
    int actor_id PK
    string actor_role
}
dim_ai_scopes {
    string scope_code PK
    string scope_description
}

```

- **Grain:** fact_laws = 1 row per law (**232 rows**). map Law requirements = 1 row per (**law × requirement_type × actor**) with an is_mandatory flag — a single law typically has many rows. map Law scopes = 1 row per (**law × scope_code**).
- **Snowflake note:** map Law requirements is a *bridge with an attribute* (is_mandatory), joining the fact to **two** dimensions at once (requirement + actor).

7.2 L1 — Controlled vocabularies (exact values)

Dimension table	Grain / PK	Values (verbatim)	Rows
dim_jurisdictions	jurisdiction_id	48 US states + European Union	49
dim_legislative_statuses	status_id	Active , Enacted , Failed/Dead , Signed	4
dim_sources	source_id	Orrick , IAPP	2
dim_actor_types	actor_id	Deployer , Developer , Provider , Distributor	4
dim_requirement_types	req_type_id	Governance Program , Assessments , Training , Responsible Individual , General Notice , Labeling/Notification , Explanation/Incident Reporting , Provider Documentation , Registration , Third-party Review , Opt-out/Appeal , Nondiscrimination	12
dim_ai_scopes	scope_code	A = all covered AI · F = foundation/frontier/general-purpose · D = automated decision-making · G = generative/synthetic · * = trained on personal data	5

7.3 L2 — Operational Postgres enums

Enum type	Values	Used by
extraction_type	obligation , definition , actor_mapping , threshold , exception , enforcement , timeline , framework_ref , ambiguity * , rights_protection , compliance_mechanism , preemption_signal	extractions.extraction_type
confidence_tier	A , B , C , D	extractions.confidence_tier

Enum type	Values	Used by
review status	pending , approved , rejected , needs_revision	extractions , bill_level_extractions , review_queue
temporal status	enacted , active , future_effective , repealed , stayed , introduced , pending , passed_one_chamber , vetoed , dead , withdrawn	document_versions.temporal_status
legalevent type	enactment , amendment , repeal , stay , effective , sunset , introduction , passage_one_chamber , veto , death , withdrawal , status_check	legal_events.event_type
dependency type	requires_definition , modifies , excepts , enforces , references , supersedes	obligation_dependencies
condition node type	AND , OR , NOT , LEAF	applicability_conditions (boolean trees)

* `ambiguity` is retained read-only for legacy rows; the standalone agent is retired (`ambiguity` now lives inline as `interpretation_risks`).

7.4 L3 — Extraction vocabularies (advisory unless marked Enforced)

Conceptual dimension	Field · agent	Allowed values (as instructed)	Enforcement
Obligation strength	<code>modality</code> · obligation	must, shall, may, should, prohibited	Advisory (<code>str</code>)
Regulated entity	<code>covered_entity_types</code> · applicability	developer, deployer, provider, operator, employer, contractor, state_agency	Advisory (<code>str[]</code>)
AI system type	<code>ai_system_types_in_scope</code> · applicability	high_risk_ai, automated_decision_system, generative_ai, facial_recognition, predictive_policing, general_purpose_ai, algorithmic_system	Advisory
Sector	<code>covered_sectors</code> · applicability	employment, housing, credit, education, healthcare, insurance, criminal_justice, financial_services, government_services, general	Advisory
Sector	<code>sector_applicability</code> · threshold	healthcare, employment, credit, housing, insurance, criminal_justice, education, government	Advisory
Penalty unit	<code>penalty_per</code> · enforcement	violation, day, occurrence	Advisory
Right type	<code>right_type</code> · rights	notice, explanation, opt_out, appeal, deletion, human_review, non_discrimination, portability, access	Advisory
Protected class	<code>protected_categories</code> · rights	consumer, employee, candidate, student, patient, minor, tenant, borrower, job_applicant	Advisory
Compliance mechanism	<code>mechanism_type</code> · compliance	impact_assessment, bias_audit, registration, certification, record_keeping, reporting, disclosure, notification	Advisory
Audit type	<code>audit_type</code> · compliance	bias_audit, impact_assessment, risk_assessment, algorithmic_audit, third_party_audit, self_certification	Advisory
Boundary class	<code>threshold_sub_type</code> · threshold	scope, temporal, exemption, other	Advisory
Consent type	<code>consent_type</code> · obligation	opt_in, opt_out, notice, notice_and_choice, disclosure	Advisory
Conflict type	<code>conflict_type</code> · preemption	federal_preemption, interstate_commerce, cross_state_conflict, first_amendment, dormant_commerce_clause, agency_jurisdiction, other	Advisory
Interpretation risk	<code>risk_type</code> · obligation, rights	vague_term, undefined_reference, conflicting_provision, scope_ambiguity, temporal_ambiguity, conditional_ambiguity	Enforced (<code>Literal</code>)
Risk severity	<code>severity</code> · obligation, rights	low, medium, high, critical	Enforced (<code>Literal</code>)

7.5 Cross-layer misalignments (the crosswalk problem)

The same legal concept is encoded by **different, non-aligned vocabularies** across layers. Any cross-layer analysis must reconcile these by hand today — there is no crosswalk table.

Concept	L1 (hard)	L3 (advisory)	Export (<code>laws_iso</code>)
Regulated actor	<code>dim_actor_types</code> — 4: Deployer, Developer, Provider, Distributor	<code>covered_entity_types</code> — 7 (adds operator/employer/contractor/state_agency, drops Distributor); <code>subject_normalized</code> — open set	tag facet <code>regulated_actor</code> : deployer, developer, provider
AI scope	<code>dim_ai_scopes</code> — 5 single-letter codes (A/F/D/G/*)	<code>ai_system_types_in_scope</code> — 7 readable strings	tag facet <code>ai_scope</code> : 4 readable strings
Lifecycle status	<code>dim_legislative_statuses</code> — 4 (Active/Enacted/Failed-Dead/Signed)	<code>temporalstatus</code> (L2) — 11	normalized 3: proposed-legislation, law-passed-not-yet-in-effect, law-in-effect
Sector	(none — no <code>dim_sectors</code>)	two divergent lists: applicability (10, incl. <code>financial_services/general</code>) vs threshold (8, uses <code>government</code>)	tag facet <code>concept</code> (overlapping but distinct)

7.6 Coverage & integrity notes

- **Jurisdiction gap:** `dim_jurisdictions` covers 48 states + EU; **Alaska (`AK`) and Ohio (`OH`) are absent.** A law from either has no FK target.
- *** wildcard scope:** `dim_ai_scopes` includes a `*` code ("trained on personal data"). Retirement is an open item (`taxonomy_strategy_summary.md` §7) — verify zero `map_law_scopes.scope_code = '*'` rows before removing.
- **Advisory drift:** because L3 is unvalidated (§7.4), expect synonyms, casing variants, and off-vocab values in `extractions.payload` / `bill_level_extractions.payload` . Profile any L3 field with `SELECT DISTINCT` before treating it as a closed set.
- **Sector has no home in L1:** the most-requested matching dimension (`covered_sectors`) exists only as advisory JSON today — no dim table, no FK. This is Phase-1/2 work in the redesign.
- **Extraction denominator:** passages are triaged (`relevant` / `not_relevant` / `uncertain`) *before* extraction; only `relevant` + `uncertain` produce records. Absence of an extraction therefore means "not found **in a triaged passage**," not "not in the law."